

SKILL WEEK 4

CREATE CUSTOMER TABLE

DISPLAY DATA

The screenshot shows the MySQL Workbench interface. On the left, the 'SCHEMAS' pane displays the database '2520080053_ruhitha'. The main editor shows a SQL script with three queries: a SELECT statement for the Customer table, a SELECT statement for the Account table, and a SELECT statement for the Bank_Transaction table. The 'Result Grid' pane displays the data for the Customer table, showing 10 rows with columns: Customer_ID, Customer_Name, Phone, Email, and City. The 'Output' pane shows the execution of the first query, indicating that 10 rows were returned.

Customer_ID	Customer_Name	Phone	Email	City
101	Ravi Kumar	9876543210	ravi@gmail.com	Hyderabad
102	Priya Sharma	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad
106	Anil Kumar	9876543215	anil@gmail.com	Delhi
107	Meena Reddy	9876543216	meena@gmail.com	Mumbai
108	Rahul Sharma	9876543217	rahul@gmail.com	Pune
109	Lakshmi Devi	9876543218	lakshmi@gmail.com	Hyderabad
110	Suresh Babu	9876543219	suresh@gmail.com	Vijayawada

The screenshot shows the MySQL Workbench interface. On the left, the 'SCHEMAS' pane displays the database '2520080053_ruhitha'. The main editor shows a SQL script with three queries: a SELECT statement for the Account table, a SELECT statement for the Customer table, and a SELECT statement for the Bank_Transaction table. The 'Result Grid' pane displays the data for the Account table, showing 10 rows with columns: Account_No, Customer_ID, Account_Type, Balance, and Branch. The 'Output' pane shows the execution of the first query, indicating that 10 rows were returned.

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	60000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
10006	106	Savings	65000.00	Delhi
10007	107	Current	150000.00	Mumbai
10008	108	Savings	35000.00	Pune
10009	109	Savings	85000.00	Hyderabad
10010	110	Current	110000.00	Vijayawada

2520080053_Ruhitha (252008005. x 2520080053_Ruhitha (25200... x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: Filter objects

2520080053_ruhitha

- Tables
- Views
- Stored Procedures
- Functions
- bookflow_db
- kludb
- sys

Query 1 SQL File 1* x SQL File 2* SQL File 3*

Limit to 1000 rows

```

109 HAVING MAX(purch_amt) > 2000;
110
111 • SELECT * FROM Customer;
112
113 • SELECT * FROM Account;
114
115 • SELECT * FROM Bank_Transaction;

```

Result Grid

	Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
12	10001	10001	DEPOSIT	10000.00	2026-08-12 15:18:28
13	10001	10001	WITHDRAW	5000.00	2026-08-12 15:18:28
14	10002	10002	DEPOSIT	15000.00	2026-08-12 15:18:28
15	10003	10003	WITHDRAW	20000.00	2026-08-12 15:18:28
16	10004	10004	DEPOSIT	5000.00	2026-08-12 15:18:28
17	10005	10005	WITHDRAW	10000.00	2026-08-12 15:18:28
18	10006	10006	DEPOSIT	12000.00	2026-08-12 15:18:28
19	10007	10007	DEPOSIT	25000.00	2026-08-12 15:18:28
20	10008	10008	WITHDRAW	5000.00	2026-08-12 15:18:28
21	10009	10009	DEPOSIT	20000.00	2026-08-12 15:18:28

Bank_Transaction 23 x

Administration Schemas

Information: Schema: 2520080053_ruhitha

Output: Action Output

#	Time	Action	Message
1	12:32:38	SELECT * FROM Customer LIMIT 0, 1000	10 row(s) returned
2	14:28:56	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned
3	14:29:14	SELECT * FROM Bank_Transaction LIMIT 0, 1000	11 row(s) returned

TRANSACTIONS

2520080053_Ruhitha (252008005. x 2520080053_Ruhitha (25200... x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: Filter objects

2520080053_ruhitha

- Tables
- Views
- Stored Procedures
- Functions
- bookflow_db
- kludb
- sys

Query 1 SQL File 1* x SQL File 2* SQL File 3*

Limit to 1000 rows

```

98 HAVING COUNT(*) >= 2;
99 • SELECT salesman_id,
100 COUNT(*) AS total_orders
101 FROM orders
102 GROUP BY salesman_id
103 HAVING COUNT(*) >= 2;
104
105 • SELECT ord_date,
106 MAX(purch_amt) AS highest_purchase
107 FROM orders
108 GROUP BY ord_date
109 HAVING MAX(purch_amt) > 2000;
110
111 • SELECT * FROM Customer;
112 • SELECT * FROM Account;
113 • SELECT * FROM Bank_Transaction;
114
115 • START TRANSACTION;

```

Output: Action Output

#	Time	Action	Message
1	12:32:38	SELECT * FROM Customer LIMIT 0, 1000	10 row(s) returned
2	14:28:56	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned
3	14:29:14	SELECT * FROM Bank_Transaction LIMIT 0, 1000	11 row(s) returned
4	17:57:52	START TRANSACTION	0 row(s) affected

DEPOSIT MONEY

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with the '2520080053_ruhitha' schema selected. The main editor shows a SQL query with the following lines:

```
102 GROUP BY salesman_id
103 HAVING COUNT(*) >= 2;
104
105 • SELECT ord_date,
106       MAX(purch_amt) AS highest_purchase
107 FROM orders
108 GROUP BY ord_date
109 HAVING MAX(purch_amt) > 2000;
110
111 • SELECT * FROM Customer;
112 • SELECT * FROM Account;
113 • SELECT * FROM Bank_Transaction;
114
115 • START TRANSACTION;
116
117 • UPDATE Account
118   SET Balance = Balance + 5000
119   WHERE Account_No = 10001;
```

The 'Output' tab at the bottom shows the execution results:

#	Time	Action	Message
✓ 1	12:32:38	SELECT * FROM Customer LIMIT 0, 1000	10 row(s) returned
✓ 2	14:28:56	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned
✓ 3	14:29:14	SELECT * FROM Bank_Transaction LIMIT 0, 1000	11 row(s) returned
✓ 4	17:57:52	START TRANSACTION	0 row(s) affected
✓ 5	17:59:16	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

CHECK BALANCE

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with the '2520080053_ruhitha' schema selected. The main editor shows a SQL query with the following lines:

```
117 • UPDATE Account
118   SET Balance = Balance + 5000
119   WHERE Account_No = 10001;
120
121 • SELECT *
122 FROM Account
123 WHERE Account_No = 10001;
```

The 'Result Grid' tab at the bottom shows the execution results:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	65000.00	Hyderabad

The 'Output' tab at the bottom shows the execution results:

#	Time	Action	Message
✓ 1	12:32:38	SELECT * FROM Customer LIMIT 0, 1000	10 row(s) returned
✓ 2	14:28:56	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned
✓ 3	14:29:14	SELECT * FROM Bank_Transaction LIMIT 0, 1000	11 row(s) returned
✓ 4	17:57:52	START TRANSACTION	0 row(s) affected
✓ 5	17:59:16	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 6	18:01:52	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned

COMMIT TRANSACTION

The screenshot shows the MySQL Workbench interface with a query editor containing a transaction. The query is as follows:

```
108 GROUP BY ord_date
109 HAVING MAX(purch_amt) > 2000;
110
111 • SELECT * FROM Customer;
112 • SELECT * FROM Account;
113 • SELECT * FROM Bank_Transaction;
114
115 • START TRANSACTION;
116
117 • UPDATE Account
118   SET Balance = Balance + 5000
119   WHERE Account_No = 10001;
120
121 • SELECT *
122   FROM Account
123   WHERE Account_No = 10001;
124
125 • COMMIT;
```

The Output tab shows the execution results:

#	Time	Action	Message
✓ 1	12:32:38	SELECT * FROM Customer LIMIT 0, 1000	10 row(s) returned
✓ 2	14:28:56	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned
✓ 3	14:29:14	SELECT * FROM Bank_Transaction LIMIT 0, 1000	11 row(s) returned
✓ 4	17:57:52	START TRANSACTION	0 row(s) affected
✓ 5	17:59:16	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 6	18:01:52	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
✓ 7	18:02:56	COMMIT	0 row(s) affected

CHECK BALANCE AFTER COMMIT

The screenshot shows the MySQL Workbench interface with a query editor containing a query to check the account balance. The query is as follows:

```
123 WHERE Account_No = 10001;
124
125 • COMMIT;
126
127 • SELECT *
128   FROM Account
129   WHERE Account_No = 10001;
```

The Result Grid shows the query results:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	65000.00	Hyderabad

The Output tab shows the execution results:

#	Time	Action	Message
✓ 2	14:28:56	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned
✓ 3	14:29:14	SELECT * FROM Bank_Transaction LIMIT 0, 1000	11 row(s) returned
✓ 4	17:57:52	START TRANSACTION	0 row(s) affected
✓ 5	17:59:16	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 6	18:01:52	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
✓ 7	18:02:56	COMMIT	0 row(s) affected
✓ 8	18:04:18	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned

ROLLBACK

WITHDRAW MONEY

2520080053_Ruhitha (252008005... x 2520080053_Ruhitha (25200... x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: 2520080053_ruhitha
Filter objects
Tables
Views
Stored Procedures
Functions
bookflow_db
kludb
sys

Query 1 SQL File 1* x SQL File 2* x SQL File 3* Limit to 1000 rows

```
116
117 • UPDATE Account
118   SET Balance = Balance + 5000
119   WHERE Account_No = 10001;
120
121 • SELECT *
122   FROM Account
123   WHERE Account_No = 10001;
124
125 • COMMIT;
126
127 • SELECT *
128   FROM Account
129   WHERE Account_No = 10001;
130
131 • UPDATE Account
132   SET Balance = Balance - 10000
133   WHERE Account_No = 10001;
```

Output

Action Output

#	Time	Action	Message
3	14:29:14	SELECT * FROM Bank_Transaction LIMIT 0, 1000	11 row(s) returned
4	17:57:52	START TRANSACTION	0 row(s) affected
5	17:59:16	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
6	18:01:52	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
7	18:02:56	COMMIT	0 row(s) affected
8	18:04:18	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
9	18:05:47	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

Schema: 2520080053_ruhitha

Object Info Session

CHECK BALANCE

2520080053_Ruhitha (252008005... x 2520080053_Ruhitha (25200... x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: 2520080053_ruhitha
Filter objects
Tables
Views
Stored Procedures
Functions
bookflow_db
kludb
sys

Query 1 SQL File 1* x SQL File 2* x SQL File 3* Limit to 1000 rows

```
132 SET Balance = Balance - 10000
133 WHERE Account_No = 10001;
134
135 • SELECT *
136   FROM Account
137   WHERE Account_No = 10001;
138
```

Result Grid

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	55000.00	Hyderabad

Account 26 x

Output

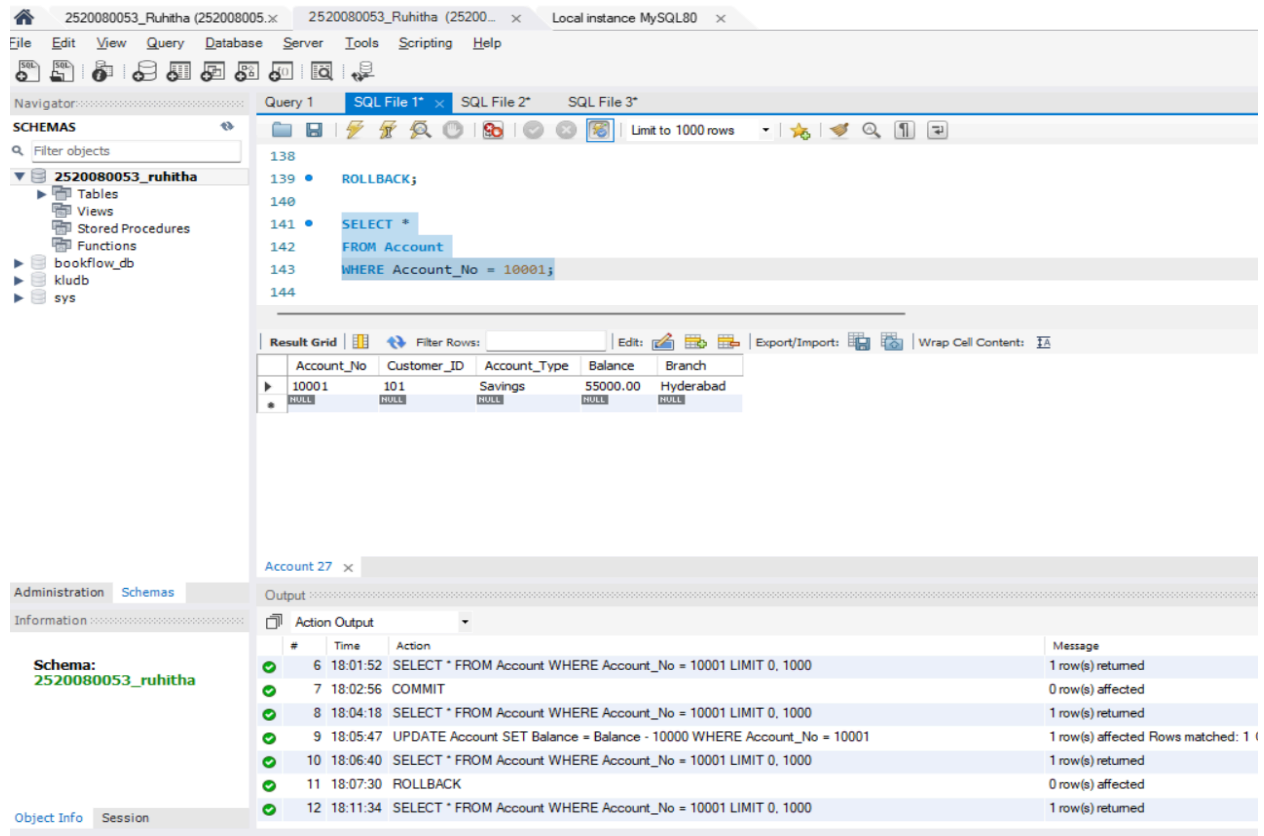
Action Output

#	Time	Action	Message
4	17:57:52	START TRANSACTION	0 row(s) affected
5	17:59:16	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
6	18:01:52	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
7	18:02:56	COMMIT	0 row(s) affected
8	18:04:18	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
9	18:05:47	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
10	18:06:40	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned

Schema: 2520080053_ruhitha

Object Info Session

CHECK BALANCE AFTER ROLLBACK



The screenshot shows the SQL Developer interface. The left sidebar displays the 'SCHEMAS' tree with the '2520080053_ruhitha' schema selected. The main query window shows a SQL script with the following lines:

```
138
139 • ROLLBACK;
140
141 • SELECT *
142 FROM Account
143 WHERE Account_No = 10001;
144
```

The 'Result Grid' shows the following data:

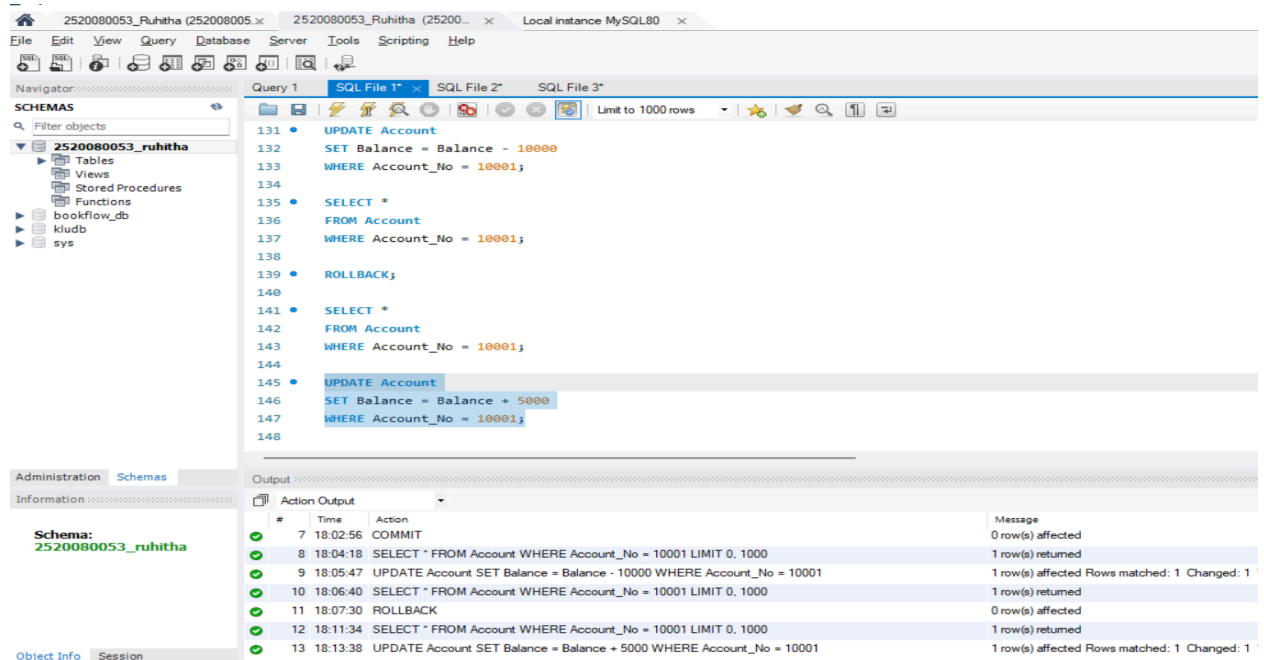
Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	55000.00	Hyderabad

The 'Output' window shows the 'Action Output' tab with the following log:

#	Time	Action	Message
6	18:01:52	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
7	18:02:56	COMMIT	0 row(s) affected
8	18:04:18	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
9	18:05:47	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1
10	18:06:40	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
11	18:07:30	ROLLBACK	0 row(s) affected
12	18:11:34	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned

SAVEPOINT

FIRST UPDATE



The screenshot shows the SQL Developer interface. The left sidebar displays the 'SCHEMAS' tree with the '2520080053_ruhitha' schema selected. The main query window shows a SQL script with the following lines:

```
131 • UPDATE Account
132 SET Balance = Balance - 10000
133 WHERE Account_No = 10001;
134
135 • SELECT *
136 FROM Account
137 WHERE Account_No = 10001;
138
139 • ROLLBACK;
140
141 • SELECT *
142 FROM Account
143 WHERE Account_No = 10001;
144
145 • UPDATE Account
146 SET Balance = Balance + 5000
147 WHERE Account_No = 10001;
148
```

The 'Output' window shows the 'Action Output' tab with the following log:

#	Time	Action	Message
7	18:02:56	COMMIT	0 row(s) affected
8	18:04:18	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
9	18:05:47	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1
10	18:06:40	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
11	18:07:30	ROLLBACK	0 row(s) affected
12	18:11:34	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
13	18:13:38	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1

CREATE SAVEPOINT

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with '2520080053_ruhitha' selected. The main editor shows a SQL script with the following lines:

```
133 WHERE Account_No = 10001;
134
135 • SELECT *
136 FROM Account
137 WHERE Account_No = 10001;
138
139 • ROLLBACK;
140
141 • SELECT *
142 FROM Account
143 WHERE Account_No = 10001;
144
145 • UPDATE Account
146 SET Balance = Balance + 5000
147 WHERE Account_No = 10001;
148
149 • SAVEPOINT Deposit1;
150
```

The 'Output' tab at the bottom shows the 'Action Output' table with the following data:

#	Time	Action	Message
8	18:04:18	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
9	18:05:47	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1
10	18:06:40	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
11	18:07:30	ROLLBACK	0 row(s) affected
12	18:11:34	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
13	18:13:38	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1
14	18:14:49	SAVEPOINT Deposit1	0 row(s) affected

SECOND UPDATE

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with '2520080053_ruhitha' selected. The main editor shows a SQL script with the following lines:

```
137 WHERE Account_No = 10001;
138
139 • ROLLBACK;
140
141 • SELECT *
142 FROM Account
143 WHERE Account_No = 10001;
144
145 • UPDATE Account
146 SET Balance = Balance + 5000
147 WHERE Account_No = 10001;
148
149 • SAVEPOINT Deposit1;
150
151 • UPDATE Account
152 SET Balance = Balance - 3000
153 WHERE Account_No = 10002;
154
```

The 'Output' tab at the bottom shows the 'Action Output' table with the following data:

#	Time	Action	Message
9	18:05:47	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1
10	18:06:40	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
11	18:07:30	ROLLBACK	0 row(s) affected
12	18:11:34	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
13	18:13:38	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1
14	18:14:49	SAVEPOINT Deposit1	0 row(s) affected
15	18:17:44	UPDATE Account SET Balance = Balance - 3000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1

CREATE SECOND SAVEPOINT

The screenshot shows the MySQL Workbench interface with the following SQL queries in the editor:

```
139 • ROLLBACK;
140
141 • SELECT *
142 FROM Account
143 WHERE Account_No = 10001;
144
145 • UPDATE Account
146 SET Balance = Balance + 5000
147 WHERE Account_No = 10001;
148
149 • SAVEPOINT Deposit1;
150
151 • UPDATE Account
152 SET Balance = Balance - 3000
153 WHERE Account_No = 10002;
154
155 • SAVEPOINT Withdrawal1;
156
```

The Output window shows the following execution results:

#	Time	Action	Message
10	18:06:40	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
11	18:07:30	ROLLBACK	0 row(s) affected
12	18:11:34	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
13	18:13:38	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1
14	18:14:49	SAVEPOINT Deposit1	0 row(s) affected
15	18:17:44	UPDATE Account SET Balance = Balance - 3000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1
16	18:19:44	SAVEPOINT Withdrawal1	0 row(s) affected

THIRD UPDATE

The screenshot shows the MySQL Workbench interface with the following SQL queries in the editor:

```
147 WHERE Account_No = 10001;
148
149 • SAVEPOINT Deposit1;
150
151 • UPDATE Account
152 SET Balance = Balance - 3000
153 WHERE Account_No = 10002;
154
155 • SAVEPOINT Withdrawal1;
156
157 • UPDATE Account
158 SET Balance = Balance + 10000
159 WHERE Account_No = 10003;
160
161
162
163
164
```

The Output window shows the following execution results:

#	Time	Action	Message
11	18:07:30	ROLLBACK	0 row(s) affected
12	18:11:34	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
13	18:13:38	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1
14	18:14:49	SAVEPOINT Deposit1	0 row(s) affected
15	18:17:44	UPDATE Account SET Balance = Balance - 3000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1
16	18:19:44	SAVEPOINT Withdrawal1	0 row(s) affected
17	18:20:31	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10003	1 row(s) affected Rows matched: 1

ROLLBACK TO SAVEPOINT

The screenshot shows the SQL Developer interface with the 'Schemas' tab selected. The '2520080053_ruhitha' schema is expanded, showing tables, views, stored procedures, and functions. The 'bookflow_db' schema is also visible. The 'sys' schema is selected. The 'Query' tab is active, showing a SQL script with the following content:

```
162
163 • COMMIT;
164
165 • SELECT *
166 FROM Account;
```

The 'Result Grid' shows the output of the query, displaying a table with columns: Account_No, Customer_ID, Account_Type, Balance, and Branch. The table contains 10 rows of data, including accounts 10001 through 10010.

The 'Output' tab is active, showing the 'Action Output' for the query. The output is as follows:

#	Time	Action	Message
14	18:14:49	SAVEPOINT Deposit1	0 row(s) affected
15	18:17:44	UPDATE Account SET Balance = Balance - 3000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1
16	18:19:44	SAVEPOINT Withdrawal1	0 row(s) affected
17	18:20:31	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10003	1 row(s) affected Rows matched: 1
18	18:22:03	ROLLBACK TO SAVEPOINT Withdrawal1	Error Code: 1305. SAVEPOINT With
19	18:25:27	COMMIT	0 row(s) affected
20	18:27:48	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned

TRANSFER MONEY

The screenshot shows the SQL Developer interface with the 'Schemas' tab selected. The '2520080053_ruhitha' schema is expanded, showing tables, views, stored procedures, and functions. The 'bookflow_db' schema is also visible. The 'sys' schema is selected. The 'Query' tab is active, showing a SQL script with the following content:

```
156
157 • UPDATE Account
158 SET Balance = Balance + 10000
159 WHERE Account_No = 10003;
160
161 • ROLLBACK TO SAVEPOINT Withdrawal1;
162
163 • COMMIT;
164
165 • SELECT *
166 FROM Account;
167
168
169 • START TRANSACTION;
170
171
172
173
```

The 'Output' tab is active, showing the 'Action Output' for the query. The output is as follows:

#	Time	Action	Message
15	18:17:44	UPDATE Account SET Balance = Balance - 3000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1
16	18:19:44	SAVEPOINT Withdrawal1	0 row(s) affected
17	18:20:31	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10003	1 row(s) affected Rows matched: 1
18	18:22:03	ROLLBACK TO SAVEPOINT Withdrawal1	Error Code: 1305. SAVEPOINT With
19	18:25:27	COMMIT	0 row(s) affected
20	18:27:48	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned
21	18:29:32	START TRANSACTION	0 row(s) affected

DEDUCT FROM SENDER

The screenshot shows the SQL Developer interface with a query window containing the following SQL code:

```
157 • UPDATE Account
158   SET Balance = Balance + 10000
159   WHERE Account_No = 10003;
160
161 • ROLLBACK TO SAVEPOINT Withdrawal1;
162
163 • COMMIT;
164
165 • SELECT *
166   FROM Account;
167
168
169 • START TRANSACTION;
170
171 • UPDATE Account
172   SET Balance = Balance - 10000
173   WHERE Account_No = 10001;
174
```

The Output window shows the execution results:

#	Time	Action	Message
✓ 16	18:19:44	SAVEPOINT Withdrawal1	0 row(s) affected
✓ 17	18:20:31	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10003	1 row(s) affected Rows matched: 1
✗ 18	18:22:03	ROLLBACK TO SAVEPOINT Withdrawal1	Error Code: 1305. SAVEPOINT With
✓ 19	18:25:27	COMMIT	0 row(s) affected
✓ 20	18:27:48	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned
✓ 21	18:29:32	START TRANSACTION	0 row(s) affected
✓ 22	18:30:44	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1

ADD TO RECEIVER

The screenshot shows the SQL Developer interface with a query window containing the following SQL code:

```
160
161 • ROLLBACK TO SAVEPOINT Withdrawal1;
162
163 • COMMIT;
164
165 • SELECT *
166   FROM Account;
167
168
169 • START TRANSACTION;
170
171 • UPDATE Account
172   SET Balance = Balance - 10000
173   WHERE Account_No = 10001;
174
175 • UPDATE Account
176   SET Balance = Balance + 10000
177   WHERE Account_No = 10002;
```

The Output window shows the execution results:

#	Time	Action	Message
✓ 17	18:20:31	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10003	1 row(s) affected Rows matched: 1 Changed: 1
✗ 18	18:22:03	ROLLBACK TO SAVEPOINT Withdrawal1	Error Code: 1305. SAVEPOINT Withdrawal1 doe
✓ 19	18:25:27	COMMIT	0 row(s) affected
✓ 20	18:27:48	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned
✓ 21	18:29:32	START TRANSACTION	0 row(s) affected
✓ 22	18:30:44	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1
✓ 23	18:32:25	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1 Changed: 1

CHECK BOTH ACCOUNTS

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with the database '2520080053_ruhitha' selected. The main editor shows a SQL query in 'Query 1':

```
177 WHERE Account_No = 10002;
178
179 • SELECT *
180 FROM Account
181 WHERE Account_No IN (10001,10002);
```

The 'Result Grid' shows the following data:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	82000.00	Vijayawada

The 'Output' pane shows the 'Action Output' table:

#	Time	Action	Message
18	18:22:03	ROLLBACK TO SAVEPOINT Withdrawal1	Error Code: 1305. SAVEPOINT W
19	18:25:27	COMMIT	0 row(s) affected
20	18:27:48	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned
21	18:29:32	START TRANSACTION	0 row(s) affected
22	18:30:44	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1
23	18:32:25	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1
24	18:33:17	SELECT * FROM Account WHERE Account_No IN (10001,10002) LIMIT 0, 1000	2 row(s) returned

COMMIT TRANSFER

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with the database '2520080053_ruhitha' selected. The main editor shows a SQL query in 'Query 1':

```
167
168
169 • START TRANSACTION;
170
171 • UPDATE Account
172 SET Balance = Balance - 10000
173 WHERE Account_No = 10001;
174
175 • UPDATE Account
176 SET Balance = Balance + 10000
177 WHERE Account_No = 10002;
178
179 • SELECT *
180 FROM Account
181 WHERE Account_No IN (10001,10002);
182
183 • COMMIT;
184
```

The 'Output' pane shows the 'Action Output' table:

#	Time	Action	Message
19	18:25:27	COMMIT	0 row(s) affected
20	18:27:48	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned
21	18:29:32	START TRANSACTION	0 row(s) affected
22	18:30:44	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1
23	18:32:25	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1
24	18:33:17	SELECT * FROM Account WHERE Account_No IN (10001,10002) LIMIT 0, 1000	2 row(s) returned
25	18:34:13	COMMIT	0 row(s) affected

FINAL BALANCES

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with the database '2520080053_ruhitha' selected. The main editor window shows a SQL query in 'Query 1':

```
187 Account_No,  
188 Balance  
189 FROM Account  
190 WHERE Account_No IN (10001,10002);  
191
```

The 'Result Grid' at the bottom displays the query results:

Account_No	Balance
10001	50000.00
10002	82000.00

The bottom status bar shows 'Account 30' and 'Output'.

DEDUCT MONEY

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with the database '2520080053_ruhitha' selected. The main editor window shows a series of SQL queries in 'Query 1':

```
178  
179 • SELECT *  
180 FROM Account  
181 WHERE Account_No IN (10001,10002);  
182  
183 • COMMIT;  
184  
185  
186 • SELECT  
187 Account_No,  
188 Balance  
189 FROM Account  
190 WHERE Account_No IN (10001,10002);  
191  
192 • UPDATE Account  
193 SET Balance = Balance - 20000  
194 WHERE Account_No = 10001;  
195
```

The 'Output' window at the bottom shows the execution results:

#	Time	Action	Message
21	18:29:32	START TRANSACTION	0 row(s) affected
22	18:30:44	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1
23	18:32:25	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1
24	18:33:17	SELECT * FROM Account WHERE Account_No IN (10001,10002) LIMIT 0, 1000	2 row(s) returned
25	18:34:13	COMMIT	0 row(s) affected
26	18:35:10	SELECT Account_No, Balance FROM Account WHERE Account_No IN (10001,10002) LIMIT 0, 1000	2 row(s) returned
27	20:28:10	UPDATE Account SET Balance = Balance - 20000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1

The bottom status bar shows 'Output' and 'Session Info'.

ADD MONEY

The screenshot shows the MySQL Workbench interface with a query window containing the following SQL code:

```
181 WHERE Account_No IN (10001,10002);
182
183 COMMIT;
184
185
186 SELECT
187     Account_No,
188     Balance
189 FROM Account
190 WHERE Account_No IN (10001,10002);
191
192 UPDATE Account
193 SET Balance = Balance - 20000
194 WHERE Account_No = 10001;
195
196 UPDATE Account
197 SET Balance = Balance + 20000
198 WHERE Account_No = 10002;
```

The Output tab shows the execution results:

#	Time	Action	Message
22	18:30:44	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1
23	18:32:25	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1
24	18:33:17	SELECT * FROM Account WHERE Account_No IN (10001,10002) LIMIT 0, 1000	2 row(s) returned
25	18:34:13	COMMIT	0 row(s) affected
26	18:35:10	SELECT Account_No, Balance FROM Account WHERE Account_No IN (10001,10002) LIMIT 0, 1000	2 row(s) returned
27	20:28:10	UPDATE Account SET Balance = Balance - 20000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1
28	20:29:08	UPDATE Account SET Balance = Balance + 20000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1

CANCEL TRANSACTION

The screenshot shows the MySQL Workbench interface with a query window containing the following SQL code:

```
183 COMMIT;
184
185
186 SELECT
187     Account_No,
188     Balance
189 FROM Account
190 WHERE Account_No IN (10001,10002);
191
192 UPDATE Account
193 SET Balance = Balance - 20000
194 WHERE Account_No = 10001;
195
196 UPDATE Account
197 SET Balance = Balance + 20000
198 WHERE Account_No = 10002;
199
200 ROLLBACK;
```

The Output tab shows the execution results:

#	Time	Action	Message
23	18:32:25	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1
24	18:33:17	SELECT * FROM Account WHERE Account_No IN (10001,10002) LIMIT 0, 1000	2 row(s) returned
25	18:34:13	COMMIT	0 row(s) affected
26	18:35:10	SELECT Account_No, Balance FROM Account WHERE Account_No IN (10001,10002) LIMIT 0, 1000	2 row(s) returned
27	20:28:10	UPDATE Account SET Balance = Balance - 20000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1
28	20:29:08	UPDATE Account SET Balance = Balance + 20000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1
29	20:29:53	ROLLBACK	0 row(s) affected

CHECK BALANCES

The screenshot shows the SQL Developer interface with a query window open. The query is as follows:

```
202 SELECT *
203 FROM Account
204 WHERE Account_No IN (10001,10002);
205
206
```

The Result Grid shows the following data:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	30000.00	Hyderabad
10002	102	Savings	102000.00	Vijayawada

The Output window shows the execution log:

#	Time	Action	Message
24	18:33:17	SELECT * FROM Account WHERE Account_No IN (10001,10002) LIMIT 0, 1000	2 row(s) returned
25	18:34:13	COMMIT	0 row(s) affected
26	18:35:10	SELECT Account_No, Balance FROM Account WHERE Account_No IN (10001,10002) LIMIT 0, 1000	2 row(s) returned
27	20:28:10	UPDATE Account SET Balance = Balance - 20000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1
28	20:29:08	UPDATE Account SET Balance = Balance + 20000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1
29	20:29:53	ROLLBACK	0 row(s) affected
30	20:32:06	SELECT * FROM Account WHERE Account_No IN (10001,10002) LIMIT 0, 1000	2 row(s) returned

ACID

ATOMICITY

The screenshot shows the SQL Developer interface with a query window open. The query is as follows:

```
203 FROM Account
204 WHERE Account_No IN (10001,10002);
205
206
207 START TRANSACTION;
208
209 UPDATE Account
210 SET Balance = Balance - 10000
211 WHERE Account_No = 10001;
212
213 UPDATE Account
214 SET Balance = Balance + 10000
215 WHERE Account_No = 10002;
216
217 COMMIT;
218
219
220
```

The Output window shows the execution log:

#	Time	Action	Message
28	20:29:08	UPDATE Account SET Balance = Balance + 20000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1
29	20:29:53	ROLLBACK	0 row(s) affected
30	20:32:06	SELECT * FROM Account WHERE Account_No IN (10001,10002) LIMIT 0, 1000	2 row(s) returned
31	20:35:09	START TRANSACTION	0 row(s) affected
32	20:35:09	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1
33	20:35:09	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1
34	20:35:09	COMMIT	0 row(s) affected

